

Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

A Clustering Framework for Parcel Co.

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Abstract

Real estate firms increasingly serve a heterogeneous population of buyers — individual home-owners, institutional investors, corporations, and international clients — each with distinct purchasing motivations and financing behavior. Treating this population as a single, undifferentiated segment leads to inefficient marketing spend, generic property recommendations, and missed investment opportunities. This paper presents an unsupervised machine learning pipeline for buyer segmentation and investment profiling, developed for Parcel Co., a real estate market-intelligence platform. Client demographic records are merged with historical property-transaction data, cleaned, and transformed through one-hot and binary feature encoding before being standardized and projected into two dimensions via Principal Component Analysis (PCA). K-Means and Agglomerative Hierarchical clustering are applied and cross-validated against each other, with the optimal number of clusters selected using the Elbow method and Silhouette analysis. A multi-criterion, optimal-assignment labeling scheme, based on the Hungarian algorithm, maps each discovered cluster to one of four business-defined buyer personas — Corporate Buyers, Luxury Investors, First-Time Buyers, and Global Investors — using investment behavior, loan usage, demographics, and geographic distribution jointly rather than a single dominant metric. The resulting segments are exposed through an interactive Streamlit dashboard supporting real-time filtering and exploration, enabling Parcel's marketing and investment teams to move from one-size-fits-all outreach toward data-driven, segment-specific strategy.

Keywords: *Customer Segmentation, K-Means Clustering, Hierarchical Clustering, Real Estate Analytics, Feature Engineering, Principal Component Analysis, Silhouette Score, Streamlit.*

1. Introduction

Real estate transactions sit at the intersection of finance, geography, and personal circumstance. A first-time home-buyer applying for a mortgage, a corporation acquiring office space, and an international investor diversifying a portfolio are all "buyers" in a transactional sense, yet they respond to entirely different marketing messages, financing options, and property recommendations. Parcel Co. operates a real estate market-intelligence platform and currently lacks a systematic, data-driven understanding of who its buyers are beyond raw transaction records. This absence of segmentation results in three concrete business problems: inefficient marketing spend directed uniformly at all clients, generic property recommendations that ignore investment intent, and missed opportunities to identify high-value investor relationships early.

Unsupervised machine learning offers a natural solution: rather than imposing a predefined segmentation scheme, clustering algorithms can discover latent buyer archetypes directly from behavioral and demographic data. This paper documents the design, implementation, and evaluation of such a pipeline, built specifically for Parcel's client and property-transaction datasets. The contributions of this work are fourfold:

- An end-to-end, reproducible data-cleaning and feature-engineering pipeline that merges client demographic records with aggregated property-transaction history.
- A comparative clustering framework using both K-Means and Agglomerative Hierarchical clustering, cross-validated against one another, with cluster count selected via the Elbow method and Silhouette analysis.
- A principled cluster-to-persona labeling mechanism that scores each cluster against four business-defined buyer profiles on investment behavior, loan usage, demographics, and geographic spread simultaneously, and resolves the cluster-to-label assignment as an optimal matching problem via the Hungarian algorithm — rather than a greedy, order-dependent heuristic.
- A deployed, interactive Streamlit dashboard that exposes segment distributions, investment behavior, geographic patterns, and per-segment insights to non-technical business stakeholders.

2. Related Work

Customer segmentation has a long history in marketing analytics, traditionally performed through rule-based frameworks such as Recency-Frequency-Monetary (RFM) scoring or fixed demographic bucketing. These approaches are

interpretable but rely on analysts pre-specifying which variables matter, and cannot discover segment structure that was not anticipated in advance.

Partitional clustering methods, most notably K-Means (MacQueen, 1967), remain the dominant unsupervised approach for segmentation tasks due to their computational efficiency and ease of interpretation. Agglomerative hierarchical clustering (Ward, 1963) offers a complementary view by building a nested hierarchy of clusters, which is useful both for validating a chosen K and for revealing sub-segment structure. Determining the appropriate number of clusters is itself a well-studied problem; this work uses the Elbow heuristic on within-cluster inertia alongside the Silhouette coefficient (Rousseeuw, 1987), which quantifies how well-separated and internally cohesive a clustering solution is, independent of the algorithm used to produce it. A broader survey of clustering methodology and its practical pitfalls is given by Jain (2010), whose observations on feature scaling and encoding sensitivity directly motivated the feature-engineering choices described in Section 5.

Applying these techniques to real estate buyer data requires care in feature representation: nominal categorical fields such as country or referral channel must be encoded so that no spurious ordinal relationship is imposed, since K-Means would otherwise treat encoded category indices as meaningfully close or far apart. This concern directly shaped the encoding strategy adopted in this work.

3. Problem Statement

Parcel currently lacks a data-driven understanding of the different types of property buyers it serves, the investment motivations that vary across demographic groups, the geographic differences in investment behavior, and customer financing patterns. In practice this leads to inefficient marketing spend, generic property recommendations that do not account for investment intent, and poor investor targeting. The objective of this work is to design an unsupervised segmentation pipeline that discovers coherent buyer groups from existing client and transaction records, interprets each group along business-relevant dimensions, and surfaces the results through a tool that Parcel's marketing and investment teams can use directly, without needing to run the underlying models themselves.

4. Dataset Description

Two datasets are used as input to the pipeline: a client dataset containing one record per buyer, and a property-transaction dataset containing one record per listing, linked to a client via a reference identifier. Table 1 describes the client-level fields, and Table 2 describes the property-transaction fields used for aggregation.

Table 1. Client dataset fields.

Field	Description
client_id	Unique client identifier
client_type	Individual / Company / other buyer entity type
gender	Gender of buyer
country	Country of residence
region	Geographic region
date_of_birth	Used to derive buyer age
acquisition_purpose	Investment / Personal use / Resale
loan_applied	Financing indicator (Yes/No)
referral_channel	Source of customer acquisition
satisfaction_score	Customer satisfaction rating

Table 2. Property-transaction dataset fields.

Field	Description
listing_id	Unique listing identifier
client_ref	Foreign key linking to client_id
sale_price	Transaction sale price

transaction_date	Date of transaction
listing_status	Sold / Active / Withdrawn
unit_category	Residential / Office / Retail / Industrial
floor_area_sqft	Unit floor area in square feet

Property records are aggregated to the client level prior to modeling: for each client, the pipeline computes the number of properties purchased, the average and total sale price across those purchases, the average floor area, and whether the client has ever purchased an office unit. Clients with no completed ("Sold") transactions are retained rather than dropped, with their aggregate fields defaulting to zero or the dataset median, since a client with no purchase history is itself a meaningful segment (e.g. a prospect who has engaged but not yet converted).

5. Methodology

5.1 Data Cleaning

Column names are normalized to lowercase snake_case, duplicate client records are removed by client_id, and free-text categorical fields are stripped of whitespace and title-cased to collapse inconsistent capitalization (e.g. "united states" and "United States" are unified). Dates are parsed with mixed-format handling to accommodate inconsistent date entry, and buyer age is derived from date of birth relative to a fixed reference date. Missing numeric fields (age, satisfaction score) are imputed with the column median, and missing categorical fields are imputed with the column mode, so that no client record is discarded purely due to partial missingness.

5.2 Feature Encoding

Two encoding strategies are applied, chosen per-field based on cardinality and semantics. Fields that are genuinely binary — gender and loan-applied status — are encoded as a single 0/1 flag, since a binary flag loses no information for a two-valued field. Nominal categorical fields with more than two possible values — client type, region, acquisition purpose, referral channel, and country — are one-hot encoded rather than label-encoded. This choice is deliberate: label encoding would assign an arbitrary integer to each category (e.g. France = 2, Germany = 5), and K-Means, being a distance-based algorithm, would then treat those integers as numerically meaningful, implicitly clustering "nearby" country codes together regardless of any real-world similarity. One-hot encoding avoids this by representing each category as an independent binary dimension. To keep the resulting feature space tractable for very high-cardinality fields, categories beyond the 15 most frequent values are grouped into an "Other" bucket before encoding.

5.3 Feature Scaling

All engineered features — demographic, behavioral, and one-hot encoded — are standardized using z-score normalization,

$$x' = (x - \mu) / \sigma$$

where μ and σ are the feature's mean and standard deviation across the client population. Standardization is necessary because the feature set mixes variables on very different scales (e.g. total investment in currency units versus a 1–5 satisfaction score); without scaling, high-magnitude features would dominate the Euclidean distance computation used by K-Means.

5.4 Dimensionality Reduction

Principal Component Analysis (PCA) is applied to the standardized feature matrix to project each client onto two principal components. This projection is used exclusively for visualization — plotting cluster assignments in two dimensions — and does not replace the full feature set used for the clustering itself.

5.5 Clustering Algorithms

Two clustering algorithms are run independently on the same standardized feature matrix. K-Means partitions clients into K clusters by iteratively minimizing within-cluster variance around K centroids, and is used as the primary segmentation algorithm due to its efficiency and straightforward interpretation. Agglomerative Hierarchical clustering builds a bottom-up hierarchy by successively merging the closest clusters (Ward linkage), and is used as a validation mechanism: the two algorithms' outputs are cross-tabulated to check whether they broadly agree on cluster structure, which increases confidence that discovered segments reflect genuine data structure rather than an artifact of one particular algorithm.

5.6 Optimal Cluster Selection

The number of clusters, K , is selected using two complementary diagnostics computed across a range of candidate K values (2 through 10). The Elbow method plots within-cluster inertia (sum of squared distances to the nearest centroid) against K and identifies the point of diminishing returns, where adding further clusters no longer meaningfully reduces inertia. The Silhouette score, defined per sample as

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

where $a(i)$ is the mean intra-cluster distance for sample i and $b(i)$ is the mean distance to the nearest neighboring cluster, provides a bounded $[-1, 1]$ measure of clustering quality that is independent of the raw inertia scale and directly comparable across different K values. Both diagnostics are surfaced in the dashboard so business users can inspect the trade-off directly rather than accepting a single fixed K .

5.7 Cluster Interpretation and Labeling

A discovered cluster index (e.g. "Cluster 2") is not directly actionable for a marketing team; it must be mapped to a named, business-meaningful persona. Parcel's requirements define four target personas: Corporate Buyers (companies purchasing multiple units), Luxury Investors (high satisfaction, large investments), First-Time Buyers (younger, loan-dependent), and Global Investors (investment-purpose buyers with broad geographic reach). Because the dataset has no direct income field, geographic diversity together with investment-purpose intent is used as the closest available proxy for the "Global Investors" profile.

For each cluster, the pipeline computes the proportion of company clients, average properties owned, average investment amount, average buyer age, loan-application rate, investment-purpose rate, average satisfaction score, and a geographic-diversity index (fraction of distinct countries represented, plus dominant country and region). Each persona is expressed as a weighted combination of these min-max normalized metrics, producing a cluster-by-persona score matrix. Rather than greedily assigning the highest-scoring persona to each cluster in sequence — which can mislabel a cluster purely due to evaluation order — the assignment is solved as a linear sum assignment problem via the Hungarian algorithm, which finds the one-to-one cluster-persona mapping that maximizes the total score across all four assignments simultaneously. When K is not exactly four, no clean one-to-one mapping exists, so each cluster is instead labeled by its single strongest-matching persona trait, with numeric suffixes to disambiguate ties.

6. System Architecture and Dashboard

The pipeline is implemented in Python using pandas for data manipulation, scikit-learn for scaling, clustering, PCA, and evaluation metrics, and SciPy for the optimal cluster-labeling assignment. Pipeline results are cached and served through an interactive Streamlit application organized into five views: a Segmentation Overview showing cluster size distribution and a PCA scatter plot comparing K-Means and Hierarchical assignments; an Investor Behavior Dashboard summarizing investment amount, loan rate, and acquisition purpose by segment; a Geographic Buyer Analysis presenting country- and region-level breakdowns including a choropleth map; a Segment Insights Panel giving descriptive statistics for a selected segment; and a Model Diagnostics view exposing the Elbow curve, Silhouette curve, feature-correlation ranking, and the underlying cluster-interpretation table described in Section 5.7. Sidebar controls allow filtering by country, region, acquisition purpose, client type, and buyer segment, and allow the number of clusters to be adjusted directly, with results cached per cluster count so repeated exploration does not require recomputation.

7. Results and Discussion

To validate the pipeline end-to-end prior to deployment on Parcel's production data, it was executed against a representative synthetic dataset of 500 clients and 900 associated property records, matching the schema in Tables 1 and 2. Absolute values below are specific to this synthetic sample and are expected to shift on Parcel's real client base, but the output structure — four labeled segments, each with a distinguishing profile — reflects what the deployed dashboard produces.

With $K = 4$, the pipeline produced an overall Silhouette score in the low positive range, consistent with real-world buyer populations where segments are meaningfully distinct but not perfectly separable — buyers rarely fall into hard-edged behavioral categories. Table 3 summarizes the resulting segment profile from this illustrative run.

Table 3. Illustrative segment profile (synthetic validation run, $K = 4$).

Segment	Dominant characteristic	Interpretation
Corporate Buyers	100% company clients; multiple properties per client	Companies acquiring multiple units, as defined in the PRD.
Luxury Investors	100% investment-purpose; highest satisfaction score	High-satisfaction clients transacting for investment purposes.

First-Time Buyers	Youngest average age; highest loan-application rate	Younger, loan-dependent buyers, as defined in the PRD.
Global Investors	Zero company share; broadest geographic spread	Individually-held accounts distributed across the widest range of countries.

Cross-tabulating K-Means against Hierarchical clustering on the same run showed substantial agreement between the two algorithms on cluster membership, which supports the conclusion that the discovered structure reflects genuine patterns in the standardized feature space rather than an artifact specific to K-Means' centroid-based approach. The feature-correlation ranking exposed in the Model Diagnostics view further confirmed that investment amount, loan status, and company-type indicators were among the strongest drivers of cluster assignment — consistent with the persona definitions the labeling scheme was designed around.

8. Limitations

- No direct income field exists in the client schema; geographic diversity combined with investment intent is used as a proxy for the "Global Investors" persona's income dimension, which may not fully capture buyer wealth.
- The reported results in Section 7 are from a synthetic validation dataset; cluster boundaries, sizes, and the specific Silhouette score will differ once run against Parcel's actual client population, and should be re-validated after deployment.
- K-Means assumes roughly spherical, similarly-sized clusters in the standardized feature space; buyer segments with highly irregular shapes may be better captured by density-based methods such as DBSCAN, which is not currently included in the comparison.
- The Hungarian-algorithm labeling scheme is only exact for $K = 4$, matching the four PRD-defined personas; for other values of K , the fallback labeling by strongest single trait is a reasonable heuristic but not a globally optimal assignment.

9. Conclusion and Future Work

This paper presented a complete, deployed pipeline for unsupervised buyer segmentation at Parcel Co., covering data cleaning, leakage-safe feature encoding, standardization, comparative K-Means and Hierarchical clustering, principled cluster count selection, and a novel optimal-assignment approach to mapping discovered clusters onto business-defined personas. The system is exposed through an interactive dashboard that lets non-technical stakeholders explore segment distributions, investment behavior, and geographic patterns without running any code directly.

Future work includes re-tuning the pipeline against Parcel's full production client base rather than the synthetic validation dataset; incorporating additional behavioral signals such as browsing or inquiry history; comparing K-Means and Hierarchical clustering against density-based alternatives such as DBSCAN and Gaussian Mixture Models; and integrating segment assignments directly into Parcel's CRM so persona labels can drive targeted campaigns automatically.

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